

Client**Server**ClientHello: $pk_{pq} \parallel pk_{tr}$ (1216 B)

encapsulate $\Rightarrow ct_{pq}, ct_{tr}, ss_{pq}, ss_{tr}$
 $K = \text{CONTEXTBOUND}(\dots)$

ServerHello: $ct_{pq} \parallel ct_{tr}$ (1120 B) $\{\text{Cert}, \text{CertVerify (ML-DSA)}, \text{Finished}\}_K$

decapsulate $\Rightarrow ss_{pq}, ss_{tr}$
 $K = \text{CONTEXTBOUND}(\dots)$; verify sig

 $\{\text{Finished}\}_K$ session established — same K unless *both* KEMs broken